

Screened multiview finite-element identification

Known geometry and measured motion → calibrated material prior → evidence-gated inverse mechanics

OBSERVATION AND RESPONSE ENCODING

1 Geometry and mechanical domain

Known surface and tetrahedral discretization
observed nodes, fixed boundary, load locations
 $\mathcal{G} = (V, T, S), \quad B$

2 Multiview deformation evidence

4 controlled loads × 3 calibrated views
learning: all 7 frames; FEM: weighted frames 1-5
 $\mathcal{Y} = \{u_{ltvi}, z_{ltvi}, w_{ltvi}, f_{lt}\}$

3 Visibility-masked response features

confidence threshold defines retained nodes
mean, peak, and temporal response; no soft weighting
 $s_i = [\bar{u}_i, \max_t u_i, \bar{w}_i, \dots]$

MATERIAL IDENTIFICATION AND PHYSICAL AUDIT

6 Acceptance, rollback, and outputs

final $\Delta < \tau$: roll back; final $\Delta \geq \tau$: accept
compact field, deformation, intervals, diagnostics
 $\hat{\theta} = (\hat{E}_b, \hat{\rho}_E, \hat{\mathbf{c}}, \hat{R}), \text{ Cl}_{90\%}$

5 Screened MAP finite-element inference

$\Delta_{probe} < \tau$: retain prior; otherwise bounded MAP FEM
OOD trigger: compare fixed-FEM and MAP physical costs
 $E_i = E_b[1 + (\rho_E - 1)o_i], \quad q \sim \mathcal{N}(1, 0.05^2)$

4 Calibrated prior and support test

train-only response head / temporal MeshGNN
validation-only fusion; train-support OOD score
 $\theta_0 = (\hat{E}_b, \hat{\rho}_E, \hat{\mathbf{c}}, \hat{R})$

Primary estimator (Ours): PCA-Ridge material prior + force-aware screened MAP FEM

Secondary representation analysis (not part of the primary estimator)

Geometry + raw static/dynamic node tokens; nodal signed-distance and partition heads; five seeds